

Parity-fixed inverse-channel normal forms and universal curvature blow-up

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Draft — July 8, 2026

Abstract

We isolate the local curvature calculus behind the complementarity law of a diagonal inverse-channel thermodynamic metric. Near a boundary where one output coordinate collapses against transverse channels, the scalar curvature has a universal Laurent normal form determined entirely by the local germ and its parity. A *generic* quadratic collapse with finite transverse channels gives an order-3 pole whose coefficient is the transverse first drift; a *parity-fixed* collapse, where the transverse channels themselves diverge and the germ is even, gives an order-4 pole with the dimension-universal coefficient $-m(m+5)/B$, where m is the transverse fibre dimension and B the collapsing amplitude—the transverse amplitudes are forgotten. For a three-dimensional state space ($m = 2$) this is $-14/B$. Kerr–Newman thermodynamic geometry is the $m = 2$ application: the extremal edge is a generic face (order 3), the two charge-reflection faces are parity-fixed (order 4, $-14/B$), and their corner is a higher-codimension blow-up stratum. All statements are checked by a reproducible symbolic certificate.

1 The local setting

Let $g = \text{diag}(g_x, g_{y_1}, \dots, g_{y_m})$ be a diagonal metric on an $(m+1)$ -dimensional state space near a face $\{x = 0\}$ across which the native coordinate x collapses. Write R for the scalar curvature. Two germs occur, distinguished by whether the transverse channels stay finite or themselves diverge, and by parity in x .

Convention. A germ is *parity-fixed* if every component is even in x (invariant under $x \mapsto -x$); then R is even in x and all odd-order poles vanish identically.

2 The pure parity-fixed normal form

Theorem 1 (Pure parity-fixed normal form). *Let $m \geq 1$, $B > 0$, $A_\alpha > 0$. For*

$$ds^2 = Bx^2 dx^2 + x^{-2} \sum_{\alpha=1}^m A_\alpha dy_\alpha^2,$$

the scalar curvature is exactly

$$R = -\frac{m(m+5)}{B} x^{-4},$$

independent of all transverse amplitudes A_α . In particular $m = 2$ gives $R = -14x^{-4}/B$.

Proof. Set $r = \frac{1}{2}\sqrt{B}x^2$, so $dr = \sqrt{B}x dx$ and $x^{-2} = \sqrt{B}/(2r)$. The metric is the warped product $ds^2 = dr^2 + w(r)^2 h_{\text{flat}}$ with flat fibre $h_{\text{flat}} = \sum_{\alpha} A_{\alpha} dy_{\alpha}^2$ and $w^2 = \sqrt{B}/(2r)$, i.e. $w \propto r^{-1/2}$. For a warped product over a flat m -dimensional fibre, $R = -2m w''/w - m(m-1)(w'/w)^2$; with $w'/w = -1/(2r)$ and $w''/w = 3/(4r^2)$,

$$R = -\frac{6m}{4r^2} - \frac{m(m-1)}{4r^2} = -\frac{m(m+5)}{4r^2},$$

and $4r^2 = Bx^4$. The two contributions are radial focusing ($6m$) and transverse fibre shear ($m(m-1)$); the amplitudes cancel because they only rescale a flat fibre, and warp curvature sees only $w'/w, w''/w$. $\square$

Corollary 2 (Asymptotic parity-fixed face). *If near $x = 0$, with all components even in x ,*

$$g_x = B(y)x^2 + O(x^4), \quad g_{y_{\alpha}} = A_{\alpha}(y)x^{-2} + O(1), \quad B, A_{\alpha} > 0,$$

then $R = -m(m+5)x^{-4}/B(y) + O(x^{-2})$. The odd pole x^{-3} is absent by parity, and the leading coefficient is independent of the amplitudes $A_{\alpha}(y)$ and of all even subleading data.

3 The generic collapse, and why order jumps by one

Proposition 3 (Generic quadratic collapse). *Let the transverse channels stay finite and the germ break parity at first order:*

$$g_x = Ax^2 + O(x^3), \quad g_{y_{\alpha}} = P_{\alpha 0} + P_{\alpha 1}x + O(x^2), \quad A, P_{\alpha 0} \neq 0.$$

Then

$$R = \frac{1}{A} \sum_{\alpha=1}^m \frac{P_{\alpha 1}}{P_{\alpha 0}} x^{-3} + O(x^{-2}) = \frac{1}{A} \sum_{\alpha} \partial_x \log P_{\alpha} \Big|_0 x^{-3} + O(x^{-2}),$$

independent of A_3 and of the quadratic transverse data $P_{\alpha 2}$.

The two propositions sit side by side and expose the mechanism. A generic face is order 3 because the odd pole is produced by the parity-breaking linear drift $P_{\alpha 1}$ of the transverse channels. A parity-fixed face has no such drift—the germ is even—so the x^{-3} term vanishes identically and the leading pole is the even x^{-4} term, whose coefficient is the dimension-universal $-m(m+5)/B$. The order is not an accident of a particular fundamental relation; it is read off the parity of the germ.

4 Higher-codimension corners

Where several parity variables vanish together the leading order is direction-dependent and is governed by a Newton polytope built from the single-face data.

Proposition 4 (Codimension-2 corner valuation). *Let two faces $\{x = 0\}$ and $\{y = 0\}$ meet at a corner, with single-face curvature orders p_x, p_y and corner-residue weights r_x, r_y (the power with which each face's leading coefficient vanishes or blows up on approach to the corner). Then near the corner R has a polar Newton polytope with vertices*

$$v_x = (-p_x, r_y), \quad v_y = (r_x, -p_y),$$

and along $x = \rho \epsilon^{a_x}, y = \epsilon^{a_y}$ the order is the support function

$$m(a) = \max_v (-\langle a, v \rangle) = \max(p_x a_x - r_y a_y, p_y a_y - r_x a_x).$$

Three consequences. (i) The order is piecewise linear—a Newton polygon—each edge carrying one face and the vertex the balanced direction, so $\min_a m(a)$ can drop *below* both single-face orders: two order-4 faces can meet at an order-2 corner. (ii) At a facet-parallel (balanced) direction the whole facet is optimal, so the leading coefficient collects the entire facet—a projective *front-face coefficient* that neither single face resolves. (iii) The face data and the raw metric weights are *equivalent* inputs, related by the explicit shift below; a single-monomial germ with the correct weights already realises the polytope. The rule is verified for synthetic (4, 4) and (4, 3) corners by direct curvature (certificate `pfc_test2`).

The vertex rule is a closed identity. Write the collapsing germ in weight form, $g_i = h_i(s) x^{p_i} y^{q_i} u_i$ with u_i a corner unit ($i = 0$ spectator, 1, 2 the two coordinates x, y).

Theorem 5 (Corner vertex rule). *The polar part of R near the corner is supported on at most the three monomials*

$$V_1 = (-(p_1 + 2), -q_1), \quad V_2 = (-p_2, -(q_2 + 2)), \quad V_0 = (-p_0, -q_0),$$

with coefficients $A(p_0, p_1, p_2)/(2h_1)$, $B(q_0, q_1, q_2)/(2h_2)$, and an s -jet C_0 , where

$$A = -p_0^2 + p_0 p_1 - p_0 p_2 + 2p_0 + p_1 p_2 - p_2^2 + 2p_2,$$

B the same in the q 's, and C_0 depends only on (h_i, h'_i, h''_i) . The corner order is the support function of the polar, nonzero-coefficient vertices. Thus $p_x = p_1 + 2$, $p_y = q_2 + 2$, $r_x = -p_2$, $r_y = -q_1$, giving $V_1 = (-p_x, r_y)$, $V_2 = (r_x, -p_y)$ of Prop. 4. A vertex disappears exactly when its coefficient vanishes ($A = 0$, $B = 0$, or a degenerate s -jet)—the precise cancellation condition—and the spectator vertex V_0 contributes only when it is polar ($p_0 > 0$ or $q_0 > 0$).

The theorem is proved by an exact scalar-curvature computation with symbolic weights (certificate `pfc_test3`); unit factors leave the vertices, hence the order, unchanged. For Kerr–Newman ($g_S \sim x^{-6}$, $g_Q \sim x^2 y^{-2}$, $g_J \sim x^{-2} y^2$) the spectator $V_0 = (6, 0)$ is non-polar, $A = -84 \neq 0$, $B = -12 \neq 0$, so the polytope is $\{V_1, V_2\} = \{(-4, 2), (2, -4)\}$ and $m(a) = \max(4a - 2, 4 - 2a)$. Codimension ≥ 3 (a Newton polytope in $\mathbb{R}^k$) and the front-face coefficient classification remain open.

5 Kerr–Newman as the $m = 2$ application

For the Kerr–Newman thermodynamic surface the inverse-channel DBP metric is diagonal in the role coordinates (S, J, Q) , a three-dimensional state space, so $m = 2$. The local types are:

divisor	local type	order	leading coefficient
$T = 0$ (extremal)	generic collapse (Prop. 3)	3	$A_2^{-1} \partial_S \log(G_J G_Q) _{\text{ext}}$
$\Omega = 0$ (spin)	parity-fixed (Cor. 2)	4	$-14/B_J$
$\Phi_e = 0$ (charge)	parity-fixed (Cor. 2)	4	$-14/B_Q$
$\Omega = \Phi_e = 0$ (Schwarzschild)	double corner (§4)	wedge	mixed stratum

The extremal order-3 and its sign, the two reflection coefficients $-14/B$, and the corner's order-2 balanced approach with its mixed excess are all established in the Kerr–Newman study; here they are exhibited as instances of the local classification, not as surface-specific miracles. The number 14 is $6 \cdot 2 + 2 \cdot 1$: radial focusing plus transverse fibre shear for a two-dimensional fibre.

Certificate

The pure normal form (Thm. 1, first-principles Ricci for $m = 1, \dots, 4$ and the symbolic-in- m warped reduction), the asymptotic parity face (Cor. 2), and the generic collapse (Prop. 3) are machine-checked in `verification/pfc_test1_local_normal_forms.py`; the codimension-2 corner valuation (Prop. 4) in `verification/pfc_test2_corner_valuation.py`; and the corner vertex rule (Thm. 5, closed symbolic identity) in `verification/pfc_test3_vertex_rule.py`. All are run twice with identical output. The Kerr-Newman instances are in `lead7_test6,7,8,9,10`.